

Project 1: Knowledge Retrieval (part - 2 Implement search algorithm)

Introduction

This part of project includes implementing Iterative Deepening Search (IDS) and A* Search algorithms to retrieve a task tree based on the user's goal, and the ability to handle multiple ways to prepare an object efficiently using heuristics and cost functions.

Features

- Load and process a FOON graph from a text file.
- Specify goal objects (e.g., dishes or foods), their states, ingredients, and containers.
- Performs both Iterative Deepening Search and A* Search through FOON nodes to find a sequence of steps required to achieve the goal.
- Saves the generated task tree to an output file.

Usage

To run the program, follow these steps:

1. **Run the main script:**

```
python test_script.py
```

Follow the prompts:

1. Enter the path to the FOON file - i.e, FOON.txt
2. Enter the goal object's name - *for example: ice*
3. Enter the goal object's state(s) (comma-separated if multiple). - *for example: crushed, frozen*
4. Enter the ingredients (comma-separated if multiple).
5. Enter the name of the container required for the goal object. – *drinking glass*

Once the process is completed, the system will search for the goal object and its matching sequence in the FOON graph. It generates an output task tree saved in a .txt file containing the recipe steps.

Output Screenshot:

For goal node i.e, ice

```
PS E:\Vignesh Ambeeru> python .\test_script.py
Enter the path to the FOON file (e.g., 'FOON.txt'): FOON.txt
-- universal foan saved to FOON.pkl
-- total functional unit: 2376
Enter the goal object name: ice
Enter the goal object state (comma-separated if multiple): crushed, frozen
Enter the goal ingredients (comma-separated if multiple):
FOON data loaded. Functional units: 2376, Object nodes: 4305
Enter the container: drinking glass
writing generated task tree to  output_IDS_ice.txt
writing generated task tree to  output_AStar_ice.txt
PS E:\Vignesh Ambeeru> 
```